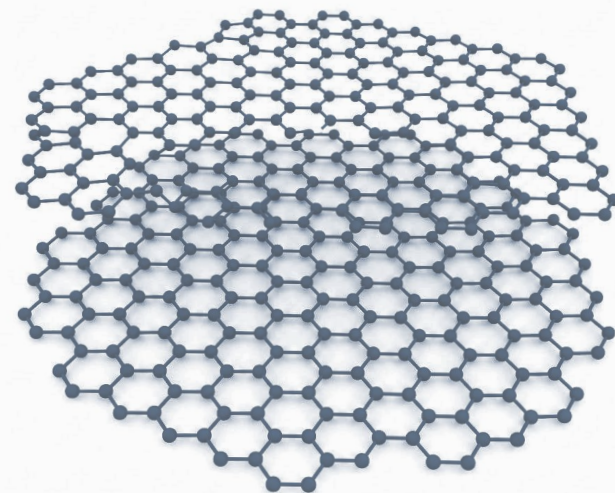


Bicapa de Grafeno Rotado:

Un superconductor impulsado por fonones

Valentina C. Barroso





Twisted Bilayer Graphene: A Phonon Driven Superconductor

Biao Lian,¹ Zhijun Wang,^{2,3} and B. Andrei Bernevig^{4,5,6}

¹*Princeton Center for Theoretical Science, Princeton University, Princeton, New Jersey 08544, USA*

²*Beijing National Laboratory for Condensed Matter Physics,
and Institute of Physics, Chinese Academy of Sciences, Beijing 100190, China*

³*University of Chinese Academy of Sciences, Beijing 100049, China*

⁴*Department of Physics, Princeton University, Princeton, New Jersey 08544, USA*

⁵*Dahlem Center for Complex Quantum Systems and Fachbereich Physik,
Freie Universitat Berlin, Arnimallee 14, 14195 Berlin, Germany*

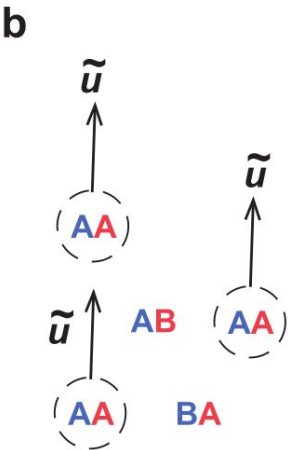
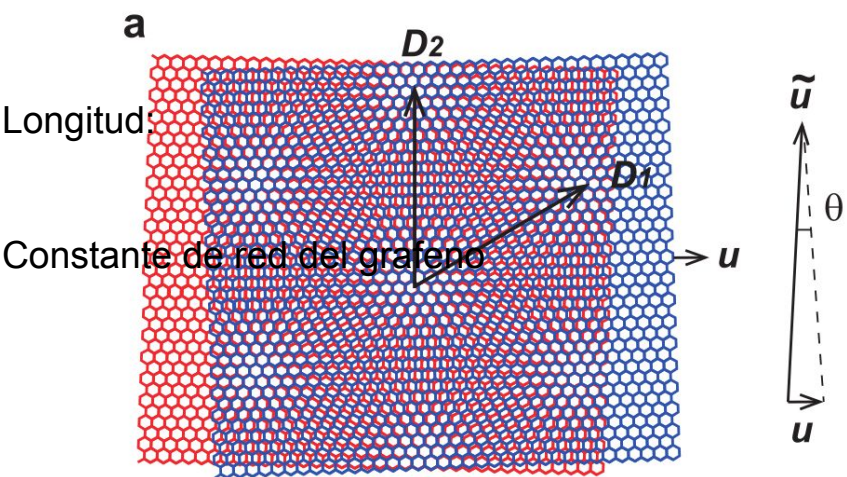
⁶*Max Planck Institute of Microstructure Physics, 06120 Halle, Germany*

(Dated: June 5, 2019)

We study the electron-phonon coupling in twisted bilayer graphene (TBG), which was recently experimentally observed to exhibit superconductivity around the magic twist angle $\theta \approx 1.05^\circ$. We show that phonon-mediated electron electron attraction at the magic angle is strong enough to induce a conventional intervalley pairing between graphene valleys K and K' with a superconducting critical temperature $T_c \sim 1K$, in agreement with the experiment. We predict that superconductivity can also be observed in TBG at many other angles θ and higher electron densities in higher Moiré bands, which may also explain the possible granular superconductivity of highly oriented pyrolytic graphite. We support our conclusions by *ab initio* calculations.

z

Superred de Moiré



$\mathbf{u}^{(j)}(\mathbf{r}) \rightarrow$ campo de fonones

$$\mathbf{u} = \mathbf{u}^{(1)} - \mathbf{u}^{(2)}$$

$$\mathbf{u}_c = (\mathbf{u}^{(1)} + \mathbf{u}^{(2)})/2$$

$$|\tilde{\mathbf{u}}| = \gamma |\mathbf{u}|$$

$$\gamma = 1/[2 \tan(\theta/2)]$$

$$\gamma \approx \theta^{-1} \gg 1$$

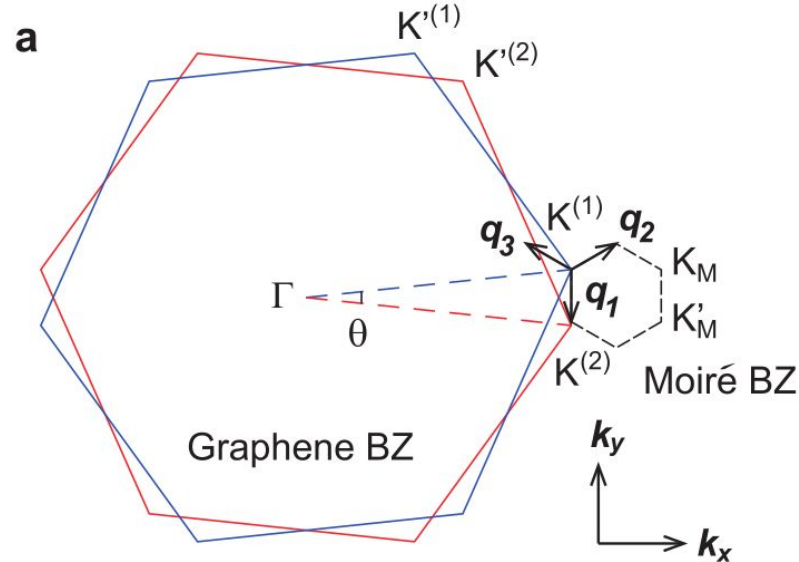
Longitud: $|D_j| = a_0/[2 \sin(\theta/2)]$

Constante de red del grafeno

$$a_0 = 0,246 \text{ nm}$$



Zona de Brillouin (BZ)

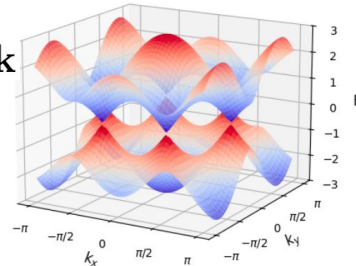


$$h^K(\mathbf{k}) = v(\sigma_x k_x - \sigma_y k_y) = \hbar v \boldsymbol{\sigma}^* \cdot \mathbf{k}$$

$$h^{K'}(\mathbf{k}) = -\hbar v \boldsymbol{\sigma} \cdot \mathbf{k}$$

$$\hbar v \approx 610 \text{ meV-nm} \text{ velocidad de fermi de grafeno}$$

$$|\mathbf{q}_j| = k_\theta = 8\pi \sin(\theta/2)/3a_0$$



Estructura del Hamiltoniano:

$$H^K(\mathbf{k}) = \begin{pmatrix} h^K(\mathbf{k}) & wT_1 & wT_2 & wT_3 \\ wT_1^\dagger & h^K(\mathbf{k}_1) & 0 & 0 \\ wT_2^\dagger & 0 & h^K(\mathbf{k}_2) & 0 \\ wT_3^\dagger & 0 & 0 & h^K(\mathbf{k}_3) \end{pmatrix}$$

$$\mathbf{k}_j = \mathbf{k} - \mathbf{q}_j$$

$$w \approx 110 \text{ meV}$$

$$T_1 = 1 + \sigma_x$$

$$T_2 = 1 - \frac{1}{2}\sigma_x - \frac{\sqrt{3}}{2}\sigma_y$$

$$T_3 = 1 - \frac{1}{2}\sigma_x + \frac{\sqrt{3}}{2}\sigma_y$$

Hamiltoniano de Dirac:

$$\tilde{H}^K(\mathbf{k}) = \frac{\langle \Psi | H^K(\mathbf{k}) | \Psi \rangle}{\langle \Psi | \Psi \rangle}$$

$$\tilde{H}^K(\mathbf{k}) = \left(\frac{1 - 3\alpha^2}{1 + 6\alpha^2} \right) \hbar v \boldsymbol{\sigma}^* \cdot \mathbf{k}$$

donde $\alpha = w/\hbar v k_\theta$

Ángulo mágico: $\theta = 1.05^\circ$ $\alpha^2 = 1/3$

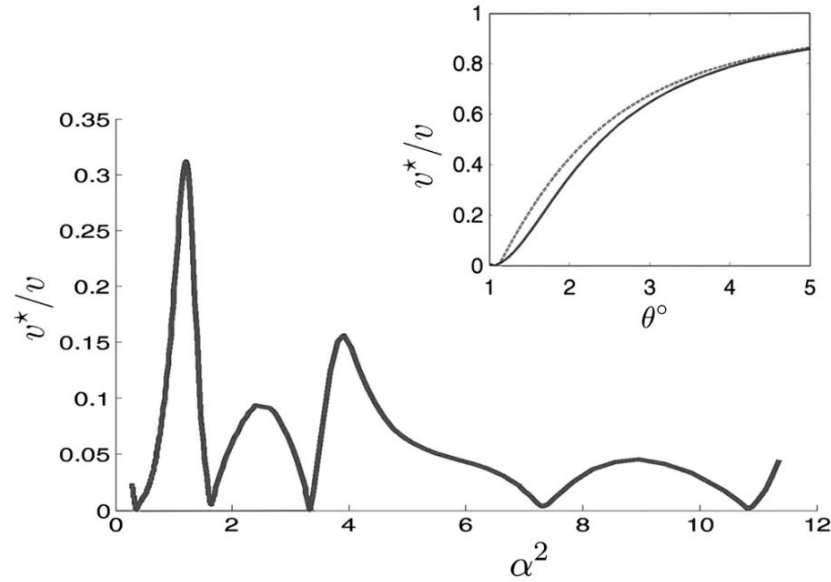


Fig. 4. Renormalized Dirac-point band velocity. The band velocity of the twisted bilayer at the Dirac point v^* is plotted vs. α^2 , where $\alpha = w/vk_\theta$ for $0.18^\circ < \theta < 1.2^\circ$. The velocity vanishes for $\theta \approx 1.05^\circ$, 0.5° , 0.35° , 0.24° , and 0.2° . (Inset) The renormalized velocity at larger twist angles. The solid line corresponds to numerical results and dashed line corresponds to analytic results based on the eight-band model.

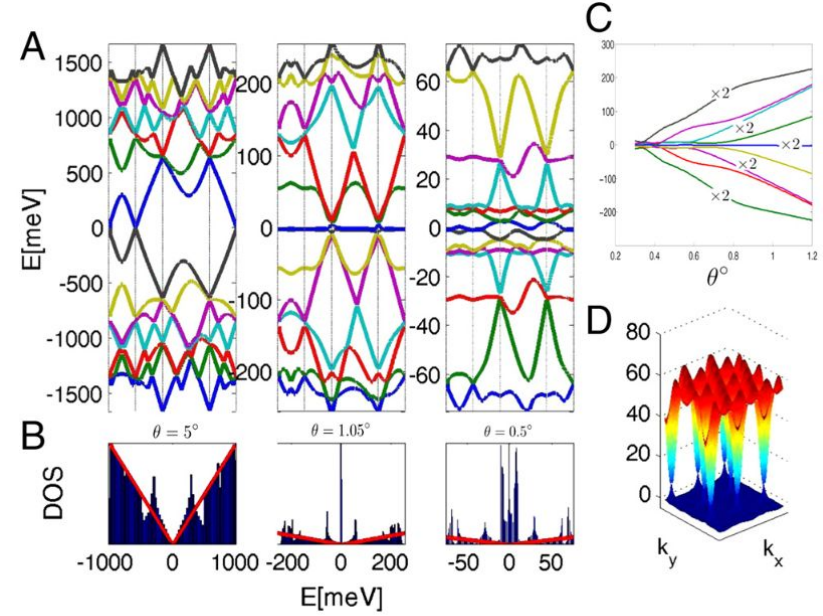
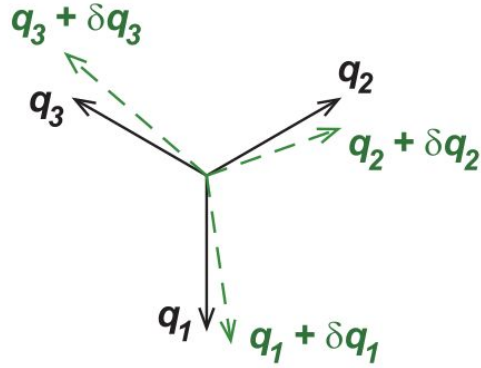


Fig. 3. Moiré bands. (A) Energy dispersion for the 14 bands closest to the Dirac point plotted along the k -space trajectory $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$ (see Fig. 1) for $w = 110$ meV, and $\theta = 5^\circ$ (Left), 1.05° (Middle), and 0.5° (Right). (B) DOS. (C) Energy as a function of twist angle for the $k = 0$ states. Band separation decreases with θ as also evident from A. (D) Full dispersion of the flat band at $\theta = 1.05^\circ$.

Rafi Bistritzer and Allan H. MacDonald, "Moiré bands in twisted double-layer graphene," Proceedings of the National Academy of Sciences 108, 12233–12237 (2011),



b



k space deformation

$\eta = \pm 1 \rightarrow$ graphene valley K, K'

$\zeta = \pm 1 \rightarrow$ Moiré BZ valley K_M, K'_M

$s = \pm 1 \rightarrow$ spin $\uparrow, \downarrow$

$$\delta \mathbf{q}_1 = \gamma k_\theta (\partial_x u_x, \partial_y u_x)$$

$$\delta \mathbf{q}_{2,3} = \gamma k_\theta \left(\pm \frac{\sqrt{3}}{2} \partial_x u_y - \frac{1}{2} \partial_x u_x, -\frac{1}{2} \partial_y u_x \pm \frac{\sqrt{3}}{2} \partial_y u_y \right)$$

$$\delta \tilde{H}(\bar{\mathbf{k}}) = H_{\text{ep}}^{\eta, \zeta, s}(\bar{\mathbf{k}}) = H_{C3}^{\eta, \zeta, s}(\bar{\mathbf{k}}) + H_{\text{SO}(2)}^{\eta, \zeta, s}(\bar{\mathbf{k}})$$

$$H_{C3}^{\eta, \zeta, s}(\bar{\mathbf{k}}) = g_{1\alpha} \eta \gamma \hbar v \psi_{\mathbf{k}'}^\dagger [\bar{k}_x (\partial_y u_x + \partial_x u_y) + \bar{k}_y (\partial_x u_x - \partial_y u_y)] \psi_{\mathbf{k}},$$

$$H_{\text{SO}(2)}^{\eta, \zeta, s}(\bar{\mathbf{k}}) = \gamma \hbar v \psi_{\mathbf{k}'}^\dagger [g_{2\alpha} (\eta \sigma_x \bar{k}_x - \sigma_y \bar{k}_y) (\partial_y u_x - \partial_x u_y) + g_{3\alpha} (\eta \sigma_x \bar{k}_y + \sigma_y \bar{k}_x) (\partial_x u_x + \partial_y u_y)] \psi_{\mathbf{k}},$$

$$g_{1\alpha} = \frac{9\alpha^2(1+3\alpha^2)}{(1+6\alpha^2)^2}$$

$$g_{2\alpha} = \frac{9\alpha^2}{(1+6\alpha^2)^2}$$

$$g_{3\alpha} = \frac{3\alpha^2}{1+6\alpha^2}$$



Hamiltoniano de Fonones

$$H_{\text{ph}} = \sum_{\mathbf{p}} \left(\hbar \omega_{\mathbf{p},L} a_{\mathbf{p},L}^{\dagger} a_{\mathbf{p},L} + \hbar \omega_{\mathbf{p},T} a_{\mathbf{p},T}^{\dagger} a_{\mathbf{p},T} \right)$$

$$\mathbf{u}(\mathbf{r}) = \sum_{\mathbf{p}} \frac{e^{i\mathbf{p} \cdot \mathbf{r}}}{\sqrt{N_s \Omega_s}} (i \hat{\mathbf{p}} u_{\mathbf{p},L} + i \hat{\mathbf{z}} \times \hat{\mathbf{p}} u_{\mathbf{p},T})$$

$$u_{\mathbf{p},\chi} = \sqrt{\frac{\hbar \Omega}{2M \omega_{\mathbf{p},\chi}}} (a_{\mathbf{p},\chi} + a_{-\mathbf{p},\chi}^{\dagger})$$

$a_{\mathbf{p},L}, a_{\mathbf{p},L}^{\dagger} \rightarrow$ operadores de fonones polarizados longitudinales

$a_{\mathbf{p},T}, a_{\mathbf{p},T}^{\dagger} \rightarrow$ operadores de fonones polarizados transversales

$$\omega_{\mathbf{p},L} = c_L p \quad \omega_{\mathbf{p},T} = c_T p$$

Hamiltoniano de interacción

$$H_{\text{int}}^{(\text{ph})} = \sum_{\mathbf{k}, \mathbf{k}'} \frac{V_{\mathbf{k}\mathbf{k}'}^{II'}(\omega)}{N_s \Omega_s} c_{\mathbf{k}',I}^{\dagger} c_{-\mathbf{k}',I'}^{\dagger} c_{-\mathbf{k},I'} c_{\mathbf{k},I}$$

Tomando $c_L = c_T \approx 10^4$ m/s. y $\alpha^2 \approx 1/3$ tenemos:

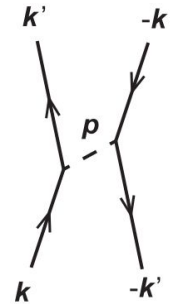
$$\frac{V_{\mathbf{k}\mathbf{k}'}^{II'}(\omega)}{\Omega_s} \approx \frac{\hbar^2 v^2 k_F^2 \varpi_{\mathbf{k}\mathbf{k}'}^{\eta\eta'}}{9M c_T^2} \frac{\omega_{\mathbf{p},T}^2}{\omega^2 - \omega_{\mathbf{p},T}^2} f_{\eta\eta'}(\varphi_{\mathbf{k}}, \varphi_{\mathbf{k}'}) \rightarrow$$

$$\mathbf{p} = \mathbf{k}' - \mathbf{k}$$

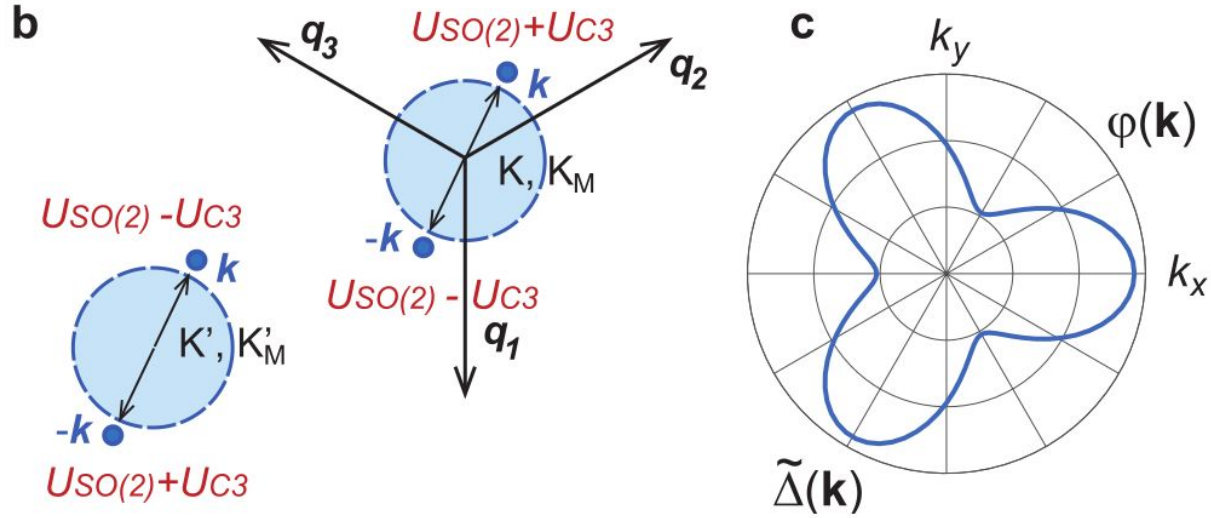
$$I = (\eta, \zeta, s)$$

$$\omega = (\xi_{\mathbf{k}'} - \xi_{\mathbf{k}})/\hbar$$

$$\xi_{\mathbf{k}} = - \left(\frac{1-3\alpha^2}{1+6\alpha^2} \right) \hbar v (|\mathbf{k}| - k_F)$$



$$f_{\eta\eta'}(\varphi_{\mathbf{k}}, \varphi_{\mathbf{k}'}) = \begin{cases} -1 - 2 \cos(\varphi_{\mathbf{k}} - \varphi_{\mathbf{k}'}), & (\eta = \eta') \\ \left| 1 - \eta \frac{e^{2i\varphi_{\mathbf{k}}} - e^{2i\varphi_{\mathbf{k}'}}}{e^{-i\varphi_{\mathbf{k}}} - e^{-i\varphi_{\mathbf{k}'}}} \right|^2, & (\eta = -\eta') \end{cases}$$



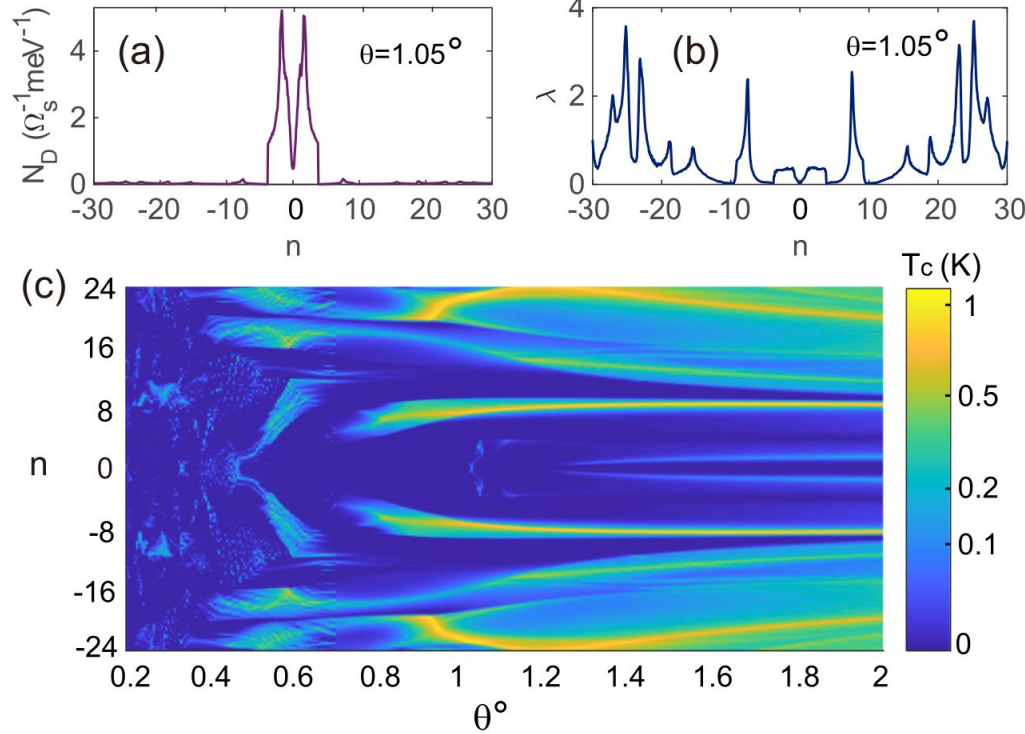
$$\Delta_k^{\eta\eta',\zeta\zeta',ss'} = \sum_{k'} V_{kk'}^{\eta\eta'} \langle c_{-k',\eta',\zeta',s'} c_{k',\eta,\zeta,s} \rangle$$

$$\Delta_k^{\eta\eta',\zeta\zeta',ss'} = -\frac{1}{2} \sum_{|\xi_{k'}| < \hbar\omega_D} \frac{V_{kk'}^{\eta\eta'} \Delta_{k'}^{\eta\eta',\zeta\zeta',ss'}}{E_{k'}^{BdG}}$$

donde $E_{k'}^{BdG} = \sqrt{|\tilde{\Delta}(k')|^2 + \xi_{k'}^2}$

Intervalley pairing amplitude

$$\Delta_{\mathbf{k}}^{\eta\eta',\zeta\zeta',ss'} = s\delta_{s,-s'}\delta_{\zeta,-\zeta'}\delta_{\eta,-\eta'}\tilde{\Delta}(\eta\mathbf{k})$$



$$N_D \gtrsim 1 \text{ meV}^{-1} \cdot \Omega_s^{-1}$$

$$\lambda \approx N_D |V_{kk'}^{II'}| \gtrsim 1$$

$$\mathcal{V}_e(q) = 2\pi e^2 / q\epsilon(q)$$

Aproximación Thomas-Fermi

$$\mathcal{V}_e(q) \approx 2\pi e^2 / \epsilon_I q_{TF} \sim N_D^{-1} \quad \left| \quad \mu_c \approx N_D \mathcal{V}_e(q) \sim 1 \right.$$

Fórmula de McMillan:

$$T_c = \frac{\hbar\omega_D}{1.45k_B} \exp \left[-\frac{1.04(1+\lambda)}{\lambda - \mu_c^*(1+0.62\lambda)} \right] \approx 0.9K$$

$$\lambda = 1.5 \text{ and } \mu_c = 1$$

$$\mu_c^* = \mu_c / [1 + \mu_c \ln(\omega_{pe}/\omega_D)]$$



$$\hbar\omega_{pe} \sim \sqrt{(4\pi n_e)^{1/2} e^2 \epsilon_F / \epsilon_I} \sim 10\hbar\omega_D$$



¡Muchas gracias por su atención!

¿Preguntas?

